

Airport Design and Simulation Lab

NAME _____ DATE _____ PERIOD _____

Fictional classroom model

These design constraints and communication steps make a tabletop simulation work. They are not FAA airport-design, phraseology, or operational-training standards.

1. Map requirements

- ☐ Two labeled, non-intersecting runways: R1 and R2
- ☐ Taxi A and Taxi B connect runways to gates
- ☐ Control tower can “see” both runways on the map
- ☐ Four labeled parking spaces or gates
- ☐ North arrow and readable route labels
- ☐ Alternate route when R1 is closed

Build route: ☐ LEGO ☐ Paper ☐ Lucid ☐ Independent map

My design responsibility: _____

2. Top-down sketch

Show runways, taxi routes, tower, four gates, north arrow, and movement arrows.



3. Predict and revise

Predicted conflict point:

First revision that could reduce it:

Readiness test: Gate 1 to R1 and R1 back to a gate.

What was blocked or confusing:

Correction before simulation:

Individual design note

Map feature and aviation role that uses it:

Conflict point:

Revision completed or still needed:

4. Five-step classroom protocol

1. **Name:** address one aircraft.
2. **Route:** state start, destination, and path.
3. **Repeat:** mover repeats the direction before moving.
4. **Confirm:** controller confirms or corrects.
5. **Log:** recorder marks result and breakdown.

Model call:

“Alpha, move from Gate 1 to the R1 hold marker by Taxi A.”

Mover repeat:

Controller confirmation or correction:

Run 1: Basic sequence

Alpha: Gate 1 to R1 hold marker. Bravo: R2 to Gate 2. Charlie: remains at Gate 3.

EVIDENCE	NOTES
Was one aircraft moving at a time?	

EVIDENCE	NOTES
Did both calls use all five steps?	
Where did the sequence or wording break down?	

Revision before Run 2:

Run 2: Two requests

Alpha requests Gate 1 to R1. Bravo requests Gate 2 to R2 at the same time. Sequence the requests and keep one aircraft at a hold marker.

Aircraft moving first: _____ Aircraft waiting: _____

Why this sequence works:

EVIDENCE	NOTES
Did the team state the order?	
Did the mover repeat each direction?	
Did the log explain why one aircraft waited?	

Revision before Run 3:

Timed revision block used: _____ minutes

What the team changed during that time:

Run 3: Runway closure

R1 is closed before movement begins. Alpha and Bravo must use an available route without crossing another moving aircraft. One gate is blocked.

Alternate route and sequence:

EVIDENCE	NOTES
Did the team use an available route?	
Did the team change the sequence?	
Did the explanation connect revision to evidence?	

Final breakdown noticed:

Final revision:

Roles used

☐ Controller ☐ Mover ☐ Recorder ☐ Safety checker ☐ Independent writer

Speaking, moving, recording, and checking are equal participation routes.

5. Individual timed iteration plan

Specific improvement goal:

Exact two- or three-minute block when the team worked on it:

Person, tool, or evidence that supported the change:

Evidence the change helped or did not help:

Next adjustment if the scenario ran again:

New individual scenario

Alpha and Bravo request movement at the same time. Taxi A is blocked.

Which aircraft moves first, and why?

Write the full five-step classroom call:

One two-minute team goal before the next run:
